CAD for VLSI

Simulated Annealing

State Space Search Problem

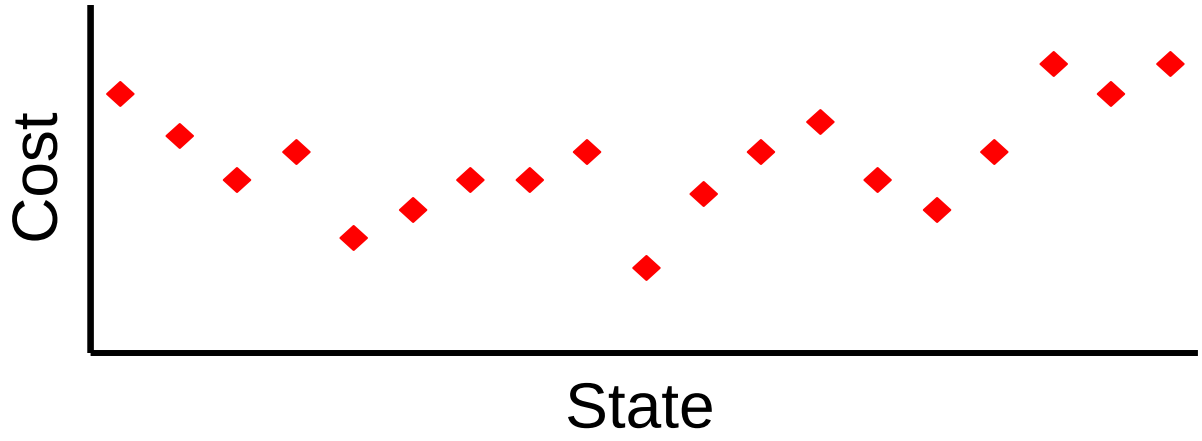
- Combinatorial optimization problems (net ordering) can be thought as a State Space Search Problem
- A State is just a configuration of the combinatorial objects involved
- The State Space is the set of all possible states (configurations)
- A Neighborhood Structure is also defined (which states can be reached in one step)
- There is a cost corresponding to each state
- Search for the min (or max) cost state

Greedy Algorithm

- A very simple technique for State Space Search Problem
- Start from any state
- Always move to a neighbor with the min cost (assume minimization problem)
- Stop when all neighbors have a higher cost than the current state

Problem with Greedy Algorithms

- Easily get stuck at local minimum
- Will obtain non-optimal solutions



- Optimal only for convex (or concave for maximization) functions



Simulated Annealing

- Very general search technique
- Try to avoid being trapped in local minimum by making probabilistic moves
- Popularize as a heuristic for optimization by:
  - Kirkpatrick, Gelatt and Vecchi, “Optimization by Simulated Annealing”, Science, 220(4598):498-516, May 1983

Basic Idea of SA

- Inspired by the *Annealing Process*:
  - The process of carefully cooling molten metals in order to obtain a good crystal structure
  - First, metal is heated to a very high temperature
  - Then slowly cooled
  - By cooling at a proper rate, atoms will have an increased chance to regain proper crystal structure
- Attaining a min cost state in simulated annealing is analogous to attaining a good crystal structure in annealing

The SA Procedure

Let *t* be the initial temperature

Repeat

Repeat

- Pick a neighbor of the current state randomly
- Let *c* = cost of current state

  Let *c'* = cost of the neighbour picked
- If *c'* < *c*, then move to the neighbour (downhill move)
- If *c'* > *c*, then move to the neighbour with probablility $e^{-(c'-c)/t}$ (uphill move)

Until equilibrium is reached

Reduce *t* according to cooling schedule

Until Freezing point is reached

Things to decide when using SA

- When solving a combinatorial problem, we have to decide:
  - The state space
  - The neighborhood structure
  - The cost function
  - The initial state
  - The initial temperature
  - The cooling schedule (how to change t)
  - The freezing point

Common Cooling Schedules

- Initial temperature, Cooling schedule, and freezing point are usually experimentally determined
- Some common cooling schedules:
  - $t = \alpha t$, where α is typically around 0.95
  - $t = e^{-\beta t}\, t$, where β is typically around 0.7
  - …

Some Other Approaches

- Genetic Algorithm
- Ant Colony Optimization
- Swarm Particle Optimization
- Tabu Search
- …